

# ETMR4115 Digital Methods

## Assignment 1

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**We were unable to upload the website, so we have attached the html-file in the delivery.**

During this assignment we have looked at different bulk carrier categories, and made examples of a sample fleet that operate between different European ports. We have used the provided port distance data to define our dimensions, trade and routes that seem realistic. Furthermore we have used these to be able to calculate different times of roundtrips, annual utilisation, which fuel consumption is relevant, and what the CO2 emissions are e.g.. At the end of the assignment EU ETS and FuelEU are discussed and how these help outline which environmental strategy for the future of the fleet can be.

You can find all the code on this github repository : <https://github.com/Basmorx/TMR4115>

## Bulk carrier categories

The description presented below of all the bulk carrier categories were found using industry standards and different official publications.

### Handymax

The handymax bulk carriers are medium-sized ships that are used in flexible trades.

Their typical size averages with:

- Deadweight: 40.000 to 50.000 tonnes
- Length: 150–200 m
- Beam: ~30 m
- Draft: ~11 m

These carriers are usually used in trades with grain, coal, and fertilizers. They are great to use in trades at smaller ports that have draft restrictions.

Its name can be split to “handy”, which indicates flexibility, and “max” that refers to the upper end of a Handy class before it moves into Supramax.

## **Panamax**

The Panamax bulk carriers are ships that are designed to be able to fit into the Panama canal.

Their typical size averages of:

- Deadweight: 65.000 to 80.000 tonnes
- Length:  $\leq 294$  m
- Beam:  $\leq 32.3$  m
- Draft:  $\sim 12.5$  m

These carriers are used in trades that include grain, coal, and bauxite on the Atlantic-Pacific routes.

Its name comes from the maximum size of vessels that are allowed in the old Panama Canal.

## **Kamsarmax**

The Kamsarmax is a vessel that is a variant of the Panamax carrier, and it is usually used in Guinea, at the port of Kamsar.

Their typical size averages of:

- Deadweight: 80.000 to 85.000 tonnes.
- Length: 229 m (which is the Kamsar port limit)
- Beam: 32 m

These vessels are usually used in trades of bauxite export from Kamsar.

Its name originates from the length restriction, the maximum length, of the port of Kamsar.

## **Capesize**

The Capesize bulk carriers are bigger vessels that exceed the ship length transit limit of the Panama and Suez canals.

Their typical size averages of:

- Deadweight: 150,000 to 200,000 tonnes
- Length:  $\sim 290$  m
- Beam:  $\sim 45$  m

The Capesize are vessels used in longer haul trades, with iron export from Brazil to China, and coal export from Australia to Europe.

Its name comes from the fact that the Capsize vessels have to sail around the Cape Horn or the Cape of Good Hope, instead of using the canals because of it exceeding the size limits.

## **VLOC**

VLOC, which stands for Very Large Ore Carrier, are extremely enormous bulk carriers that are designed for quite big long haul iron ore transports.

Their typical size averages of:

- Deadweight: 200,000 to 400,000 tonnes.
- Length: ~360 m
- Beam: 60-65 m

Their trade routes are usually exporting iron ore between ports in Brazil and China.

Its name “Very Large Ore Carrier” is based on the fact that these vessels are among one of the largest ore carriers in the shipment world.

## **Fleet definition and ports**

By using the provided distances.json file, our group have selected these European and EEA ports that are identified by their LOCODEs:

- **BEANR**: Antwerp (Belgium)
- **NLRTM**: Rotterdam (Netherlands)
- **GBABD**: Aberdeen (UK)
- **SEGOT**: Gothenburg (Sweden)
- **NOKRS**: Kristiansand (Norway)
- **SEGVX**: Gavle (Sweden)

These ports are used to form the basis of our roundtrip routes in our sample fleet.

## Sample vessels and routes

We used data from representative vessels in each category to determine key metrics, including deadweight tonnage (DWT), average speed, annual CO<sub>2</sub> emissions, and CO<sub>2</sub> equivalent per nautical mile. These figures were sourced from the EMSA and VesselFinder databases. Every data can be verified by using the IMO number.

For the Handymax vessel :

- IMO : 9703411
- 231.6 kg CO<sub>2</sub>eq / n m
- DWT : 40799 T
- Average speed : 10.5 kn
- CO<sub>2</sub> emission per year : 4327.35 T

For the Panamax vessel :

- IMO : 9638927
- 260.14 kg CO<sub>2</sub>eq / n m
- DWT : 52713 T
- Average speed : 9.4 kn
- CO<sub>2</sub> emission per year : 2563.65 T

For the Kamsarmax vessel :

- IMO : 9724776
- 305.86 kg CO<sub>2</sub>eq / n m
- DWT : 63073 T
- Average speed : 11.1 kn
- CO<sub>2</sub> emission per year : 5324.99 T

For the Capesize vessel :

- IMO : 9839131
- 1083.86 kg CO<sub>2</sub>eq / n m
- DWT : 220766 T
- Average speed : 15.8 kn
- CO<sub>2</sub> emission per year : 89078.71 T

For VLOC :

- IMO : 9488918
- 815 kg CO<sub>2</sub>eq / n m
- DWT : 402 347 T
- Average speed : 15.4 kn
- CO<sub>2</sub> emission per year : 4327.35 T

## Routes and calculations

For each of our bulk carrier categories, we have to define one representative vessel and a simplified roundtrip route between a few of our selected ports.

We will count a 10h stop at each port during the roundtrip.

- **Handymax**

The chosen route is : Port of Aarhus (DKAAR) -> Port of Aberdeen (GBABD) -> Port of Rotterdam (NLRTM) -> Port of Aarhus (DKAAR)

The total distance is 1521 nm. To calculate CO<sub>2</sub> emissions for the voyage, we used the CO<sub>2</sub>eq emissions per nautical mile sourced from the EMSA website.

This voyage emitted 352 263.6 kg of CO<sub>2</sub>eq, giving us a CII score of 5.68 gCO<sub>2</sub>/dwt · nm. We calculated this score using the formula: (Total voyage emissions) / (capacity × distance).

The travel time for this roundtrip is 144.9h + 20h = 164.9h.

We have implemented this formula inside a python script for the next vessels.

- **Panamax**

The chosen route: Port of Gävle (SEGVX) -> Port of Antwerp (BEANR) -> Port of Kokta (FIKTK) -> Port of Gävle (SEGVX).

The total distance is 2880 nautical miles and this voyage emitted 749 203.2 kg of CO<sub>2</sub>eq, giving us a CII score of 4.935 gCO<sub>2</sub>/dwt · nm.

The travel time for this roundtrip is 306.4h + 20h = 326.4h.

- **Kamsarmax**

The chosen route: Port of Rotterdam (NLRTM) -> Port of Kristiansand (NOKRS) -> Port of Gothenburg (SEGOT) -> Port of Rotterdam (NLRTM).

The total distance is 1050 nautical miles and this voyage emitted 321 153 kg of CO<sub>2</sub>eq, giving us a CII score of 4.85 gCO<sub>2</sub>/dwt · nm.

The travel time for this roundtrip is 94.6h + 20h = 114.6h.

- **Capesize**

The chosen route: Port of Antwerp (BEANR) -> Port of Gävle (SEGVX) -> Port of Gothenburg (SEGOT) -> Port of Antwerp (BEANR).

The total distance is 2489 nautical miles and this voyage emitted 2 697 727.5 kg of CO<sub>2</sub>eq, giving us a CII score of 4.91 gCO<sub>2</sub>/dwt · nm.

The travel time for this roundtrip is 157.5h + 20h = 177.5h.

- **VLOC**

The chosen route: Port of Antwerp (BEANR) -> Port of Kokta (FIKTK) -> Port of Gothenburg (SEGOT) -> Port of Antwerp (BEANR).

The total distance is 2647 nautical miles and this voyage emitted 2 157 305 kg of CO<sub>2</sub>eq, giving us a CII score of 2.03 gCO<sub>2</sub>/dwt · nm.

The travel time for this roundtrip is 171.9h + 20h = 191.9h.

## **CII ratings**

The attained CII values used in these ratings are calculated by using the IMO formula:  $CII = \text{Annual CO}_2 \text{ emissions (g)} / (\text{Capacity} \times D \times \text{correction factors})$

To determine the operational CII rating (A-E), we compare the attained CII with the required CII value for the vessel type. Since the assignment does not provide required CII values, we chose reasonable placeholder values that were inspired by typical IMO reference lines for bulk carriers: We then applied a simple rating scale:

- **A:**  $\text{Attained} \leq 0.75 \times \text{Required}$
- **B:**  $0.75 - 0.95 \times \text{Required}$
- **C:**  $0.95 - 1.05 \times \text{Required}$
- **D:**  $1.05 - 1.20 \times \text{Required}$
- **E:**  $> 1.20 \times \text{Required}$

For this assignment, we used a reduction factor of  $z = 5\%$ , giving:  $\text{Required CII} = (1 - 0.05) \times CII_R$

This gives the following required CII values:

| Vessel    | Required CII (Placeholder) | Required CII (z = 5%) |
|-----------|----------------------------|-----------------------|
| Handymax  | 5.50                       | 5.225                 |
| Panamax   | 4.80                       | 4.56                  |
| Kamsarmax | 4.80                       | 4.56                  |
| Capesize  | 4.00                       | 3.80                  |
| VLOC      | 3.20                       | 3.04                  |

We then compare each vessel's attained CII with its required CII:

- Handymax:  $5.68/5.225 \approx 1.087 \rightarrow \mathbf{D}$
- Panamax:  $4.935/4.56 \approx 1.087 \rightarrow \mathbf{D}$
- Kamsarmax:  $4.85/4.56 \approx 1.064 \rightarrow \mathbf{D}$
- Capesize:  $4.91/3.80 \approx 1.292 \rightarrow \mathbf{E}$
- VLOC:  $2.03/3.04 \approx 0.667 \rightarrow \mathbf{A}$

Final ratings:

| Vessel    | Attained CII | Required CII | Rating |
|-----------|--------------|--------------|--------|
| Handymax  | 5.68         | 5.225        | D      |
| Panamax   | 4.935        | 4.56         | D      |
| Kamsarmax | 4.85         | 4.56         | D      |
| Capesize  | 4.91         | 3.80         | E      |
| VLOC      | 2.03         | 3.04         | A      |

## EEXI/EEDI

The calculations of EEXI/EEDI based on reference speed  $V_{ref}$ , installed power and design parameters are included here.

To calculate the EEDI/EEXI index, our group has decided to use these reference speeds (in kn) and installed power (in kW) as there were not provided any numbers in the assignment for these calculations:

- Handymax: 13 kn & 8,000 kW

- Panamax: 14 kn & 10,000 kW
- Kamsarmax: 14 kn & 11,000 kW
- Capesize: 15 kn & 20,000 kW
- VLOC 15 kn & 25,000 kW

Furthermore, we have decided to use HFO as the fuel type for all vessels.

In the lecture, we were presented this EEXI formula:  $EEXI = CO_2 / (Capacity * V_{ref})$

By using the already calculated CO<sub>2</sub> values, we obtain these EEXI values:

| Vessel    | EEXI (gCO <sub>2</sub> /(t.kn)) |
|-----------|---------------------------------|
| Handymax  | 524.202                         |
| Panamax   | 688,922                         |
| Kamsarmax | 279,75                          |
| Capesize  | 999.53                          |
| VLOC      | 479,401                         |

## EU ETS Compliance

Based on the lecture slides, the ships must surrender to the EU Allowances (EUAs) that corresponds to a share of their CO<sub>2</sub> emissions:

- 2024: 40%
- 2025: 70%
- 2026: 100%

We are to assume that all the voyages are intra-EU and that by 2026 the emissions are to be 100% covered. With using an EUA price of 80 €/t CO<sub>2</sub> (for 2026), we can estimate the annual ETS costs of each vessel:

| Vessel   | Annual CO <sub>2</sub> (t) | EUAs needed (t) | Dummy cost at 80 €/t |
|----------|----------------------------|-----------------|----------------------|
| Handymax | 352,3                      | 352,3           | ≈ 28 181,04 €        |

|                  |         |          |                |
|------------------|---------|----------|----------------|
| <b>Panamax</b>   | 749,3   | 749,3    | ≈ 59 944,256 € |
| <b>Kamsarmax</b> | 23,800  | 321, 153 | ≈ 25 692,24 €  |
| <b>Capesize</b>  | 2 697,7 | 2 697,7  | ≈ 215 818,2 €  |
| <b>VLOC</b>      | 2 157,3 | 2 157,3  | ≈ 172 584,4 €  |

For 2024, 65€/tCO<sub>2</sub>

| <b>Vessel</b>    | <b>Annual CO<sub>2</sub> (t)</b> | <b>EUAs needed (t)</b> | <b>Dummy cost at 65 €/t</b> |
|------------------|----------------------------------|------------------------|-----------------------------|
| <b>Handymax</b>  | 352,3                            | 352,3                  | ≈ 9 158,838 €               |
| <b>Panamax</b>   | 749,3                            | 749,3                  | ≈ 19 481,883 €              |
| <b>Kamsarmax</b> | 23,800                           | 321, 153               | ≈ 8 349,978 €               |
| <b>Capesize</b>  | 2 697,7                          | 2 697,7                | ≈ 70 140,915 €              |
| <b>VLOC</b>      | 2 157,3                          | 2 157,3                | ≈ 56 089,93 €               |

For 2025, 73€/tCO<sub>2</sub>

| <b>Vessel</b>    | <b>Annual CO<sub>2</sub> (t)</b> | <b>EUAs needed (t)</b> | <b>Dummy cost at 73 €/t</b> |
|------------------|----------------------------------|------------------------|-----------------------------|
| <b>Handymax</b>  | 352,3                            | 352,3                  | ≈ 18 106,67 €               |
| <b>Panamax</b>   | 749,3                            | 749,3                  | ≈ 38 514,934 €              |
| <b>Kamsarmax</b> | 23,800                           | 321, 153               | ≈ 16 507,585 €              |
| <b>Capesize</b>  | 2 697,7                          | 2 697,7                | ≈ 138 665,891 €             |
| <b>VLOC</b>      | 2 157,3                          | 2 157,3                | ≈ 110 887,634 €             |

## FuelEU Compliance

FuelEU Maritime sets limits on how much greenhouse gas (GHG) energy ships can use when they operate in the EU. The rules look at the ship's overall energy use from production to combustion and compare it to a 2020 baseline (DNV, 2026). Each vessel receives a compliance balance, which shows whether its GHG intensity is below or above the allowed limit for the given year.

Since the assignment doesn't specify a fuel type, we chose to assume that all vessels run on conventional HFO. With this fuel, the GHG intensity for each vessel ends up above the FuelEU limits for the relevant compliance years. This means that, in their current state, none of the vessels meet the FuelEU requirements, and it results in a negative compliance balance, meaning each vessel has a deficit and is therefore non-compliant.

The table below shows the calculated energy use, the compliance balance, and the resulting penalty for each vessel:

| Vessel    | Energy used (MJ) | Compliance Balance (gCO <sub>2</sub> eq) | Status  | Penalty (€) |
|-----------|------------------|------------------------------------------|---------|-------------|
| Handymax  | 3,839,629.84     | -9,241,989.02                            | Deficit | 6,055.66    |
| Panamax   | 8,167,324.29     | -19,658,749.58                           | Deficit | 12,881.07   |
| Kamsarmax | 3,500,534.09     | -8,425,785.57                            | Deficit | 5,520.86    |
| Capesize  | 29,404,942.01    | -70,777,695.42                           | Deficit | 46,375.91   |
| VLOC      | 23,514,398.76    | -56,599,157.82                           | Deficit | 37,085.66   |

Because all vessels show a deficit, none of them meet FuelEU requirements in their current configuration. To become compliant, the fleet would need to make changes such as switching to cleaner fuels (such as biofuel blends, e.g.), using shore power while at berth, or joining a pooling arrangement where operators can share compliance balances. Without these kinds of changes, a fleet running purely on HFO will not meet FuelEU's long-term targets.

## **Environmental strategy**

2030

By 2030, our short-term goal is to reduce emissions through operational measures rather than vessel changes. This includes sailing at reduced speed where schedules allow, keeping hulls clean, and using small biofuel blends. We also plan to connect to shore power at EU ports where available. Based on our calculations, Capesize currently has the lowest CII rating and the largest FuelEU deficit, so these measures target that vessel first.

2040

By 2040, we plan to start replacing our oldest vessels, prioritizing those that fail to meet the tightened EEXI limits, with newer vessels designed to lower EEXI/EEDI values. We also plan to begin introducing low-carbon fuels on some routes, moving beyond the biofuel blends used in 2030.

2050

By 2025, our goal is to meet the FuelEU Maritime target of an 80% reduction in GHG intensity relative to the 202 baseline (DNV, 2026). This requires the fleet to run substantially on alternative fuels rather than HFO, with all vessels targeting a rating of A or B.

## **Conclusion**

In this assignment, we put together a small example fleet of bulk carriers and chose realistic routes between different European ports using the distance data we were given. We calculated roundtrip times, the use of fuel, CO<sub>2</sub> emissions and carbon intensity for each of the vessels to get a better understanding of how they perform on these routes.

When we looked at the results together with the CII, EEXI/EEDI, EU ETS and FuelEU framework, we noticed a clear pattern. Larger ships tend to have lower carbon intensity, but

they also produce more total emissions and therefore face higher ETS costs. Smaller ships pollute less overall, but they are not as efficient when you look at emissions per transported tonne.

## Sources

DNV. (2026). *What is FuelEU Maritime?* DNV. Retrieved August 26th, 2026, from

<https://www.dnv.com/maritime/insights/topics/fueleu-maritime/>

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<https://www.vesselfinder.com/vessels/>

NTNU. (2026). distances.json

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## Usage of AI

In this assignment, we have used artificial intelligence as support for our work by using it to help us create the HTML layout for the website and as a programming helper for the python code. AI was also used to check grammar and the structure of the paragraphs. AI did not perform any of the required calculations for us, but helped with formatting and wording so it was easier to present. We have to make clear that all calculations and methodical choices were made by our group, and we kept full control over the presentation of our numbers, assumptions and final results.